

Public Finance: Economic Impact Analysis of Government



Edoardo Cefala



Inefficiencies of Taxation



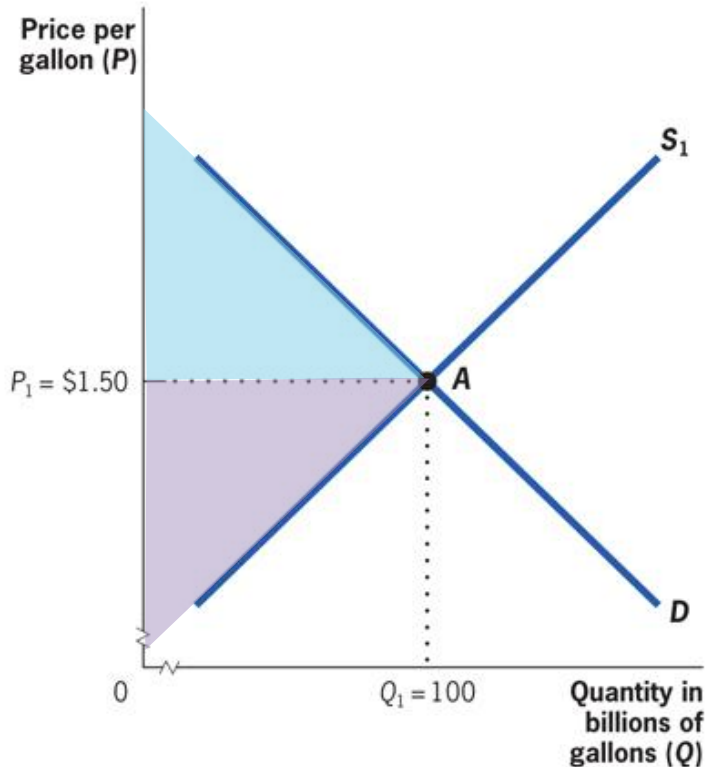
Chapter 20 (1)

We are moving on from the effects of taxes on prices to the effects of taxes on quantities.

-> What effect does a tax have on social welfare?

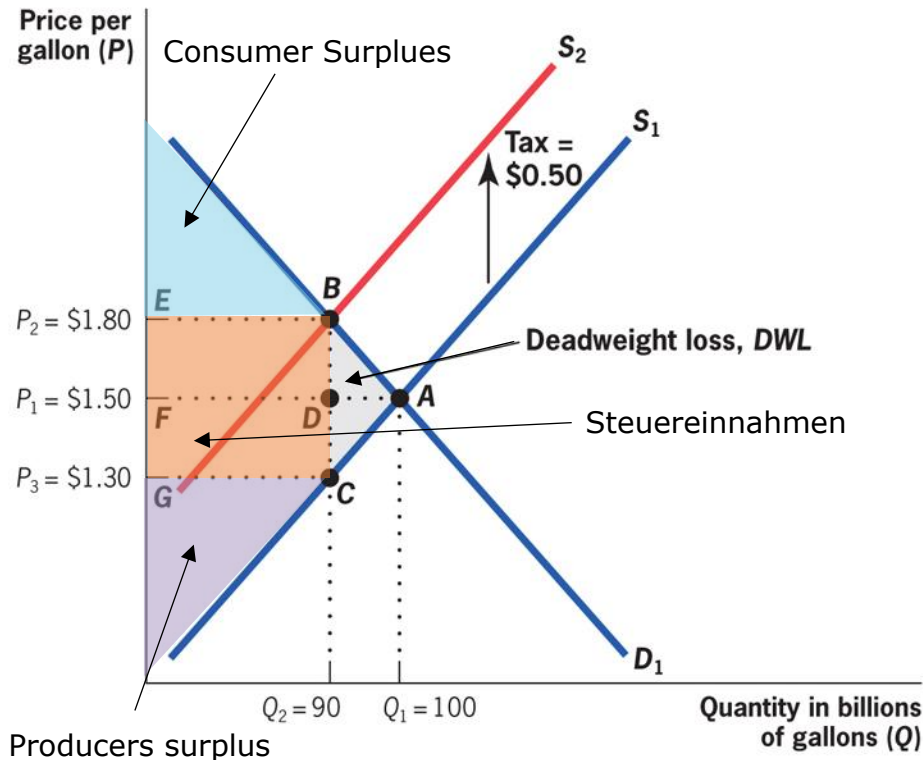
Consumer and Producer Surplus

- Consumer Surplus: $WTP > Price$
- Producer Surplus: $Price > Marginal Costs$

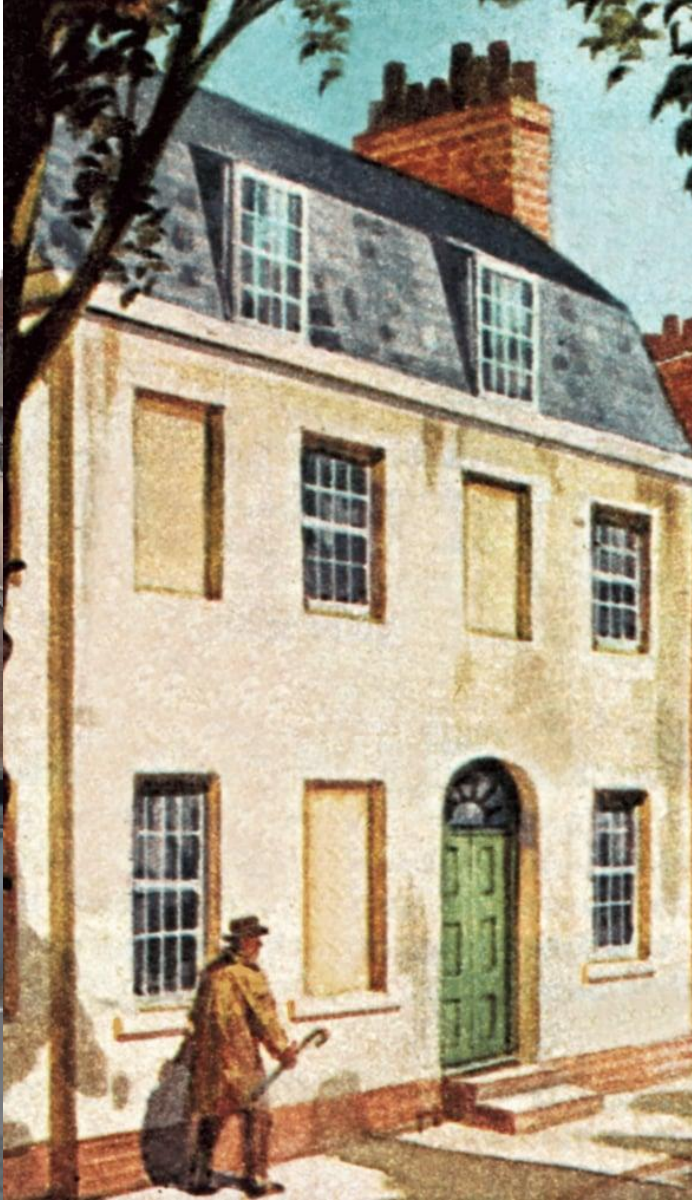


- The demand curve represents the Willingness to Pay (WTP) of consumers.
- The supply curve represents the marginal costs of production.
- All units consumed up to the equilibrium quantity generate surplus.
- These are maximized at equilibrium.

Deadweight loss



- Through the tax on producers, the equilibrium shifts from A to B .
- The quantity traded decreases and not all (potential) trades with a positive surplus are carried out.
- This reduces consumer surplus by $EBAF$ and producer surplus by $FACG$.
- Tax revenue = $EBCG$
- Deadweight loss = BAC -> What is lost through the introduction of the tax and is not returned in the form of tax revenue.
- Assumption that tax revenue creates neither more nor less social benefit than its monetary value.



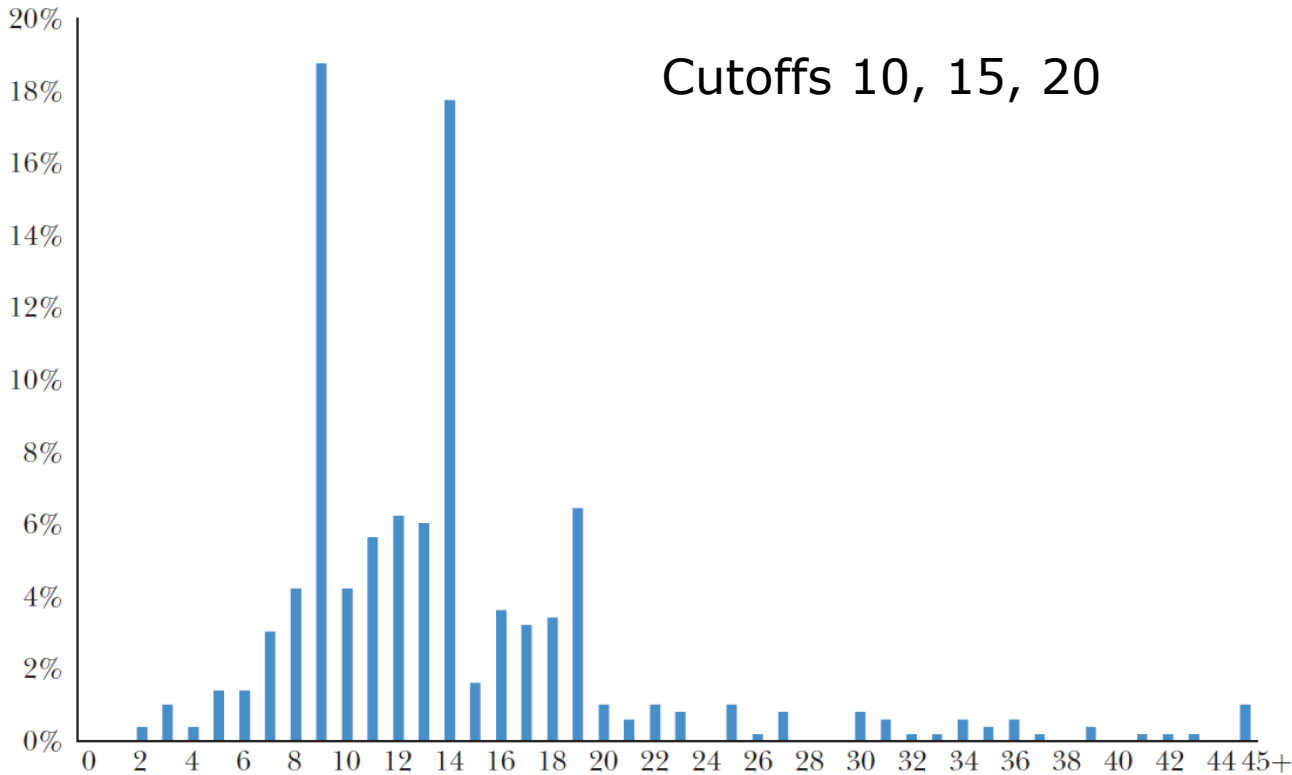
Welfare Loss (Deadweight Loss - DWL) Examples of Tax Evasion from History - Window Tax

- Tax on the number of windows of a house
- As a proxy for wealth 17th-19th century in some European countries (e.g., FR, DE, UK,...)
- Oates, W. E. and Schwab, R. M. (2015) examine behavioral effects of the window tax in England
- The amount of the tax rate depends on the number of windows:
 - A house with 10-14 windows pays 6 pence per window
 - A house with 15-19 windows pays 9 pence per window
- Does the number of windows adapt to the tax brackets?

Welfare Loss (Deadweight Loss - DWL)

Examples of Tax Evasion from History - Window Tax

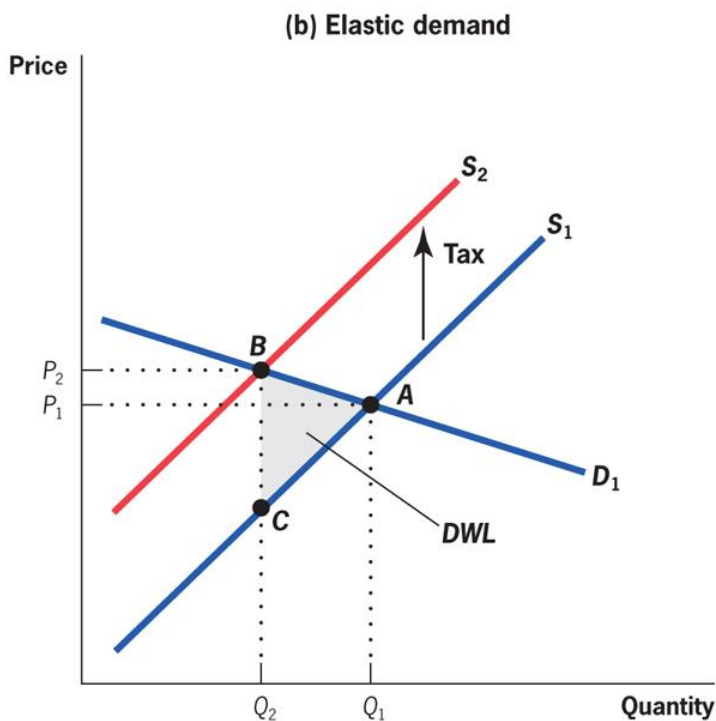
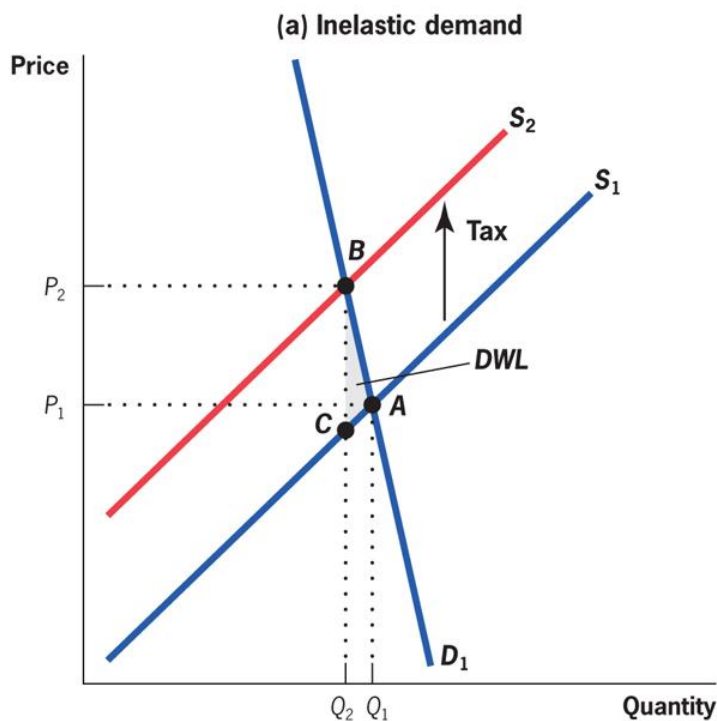
Distribution of Number of Windows, 1747–1757 Sample



Quelle: Oates and Schwab (2015)

Welfare Loss (Deadweight Loss) and Elasticity

- **Inelastic Demand:** Tax leads to a large price change, quantity barely changes
- **Elastic Demand:** Tax leads to a small price change, quantity reacts strongly



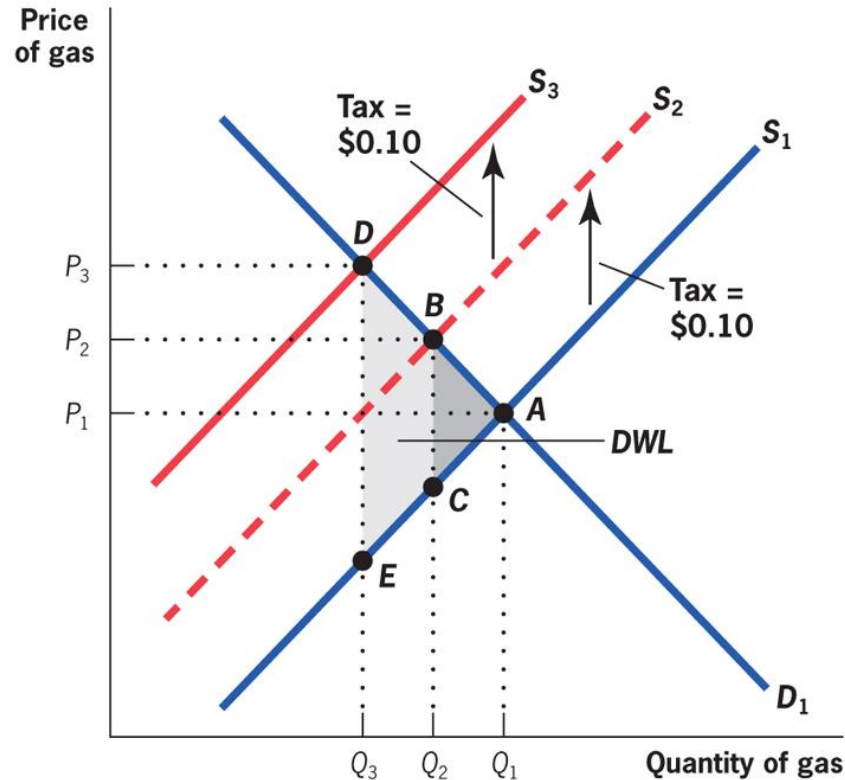
Welfare Loss (Deadweight Loss) and Elasticity

- **Elasticity determines...**
 - Distribution of the tax burden
 - Size and distribution of the welfare loss

 - **Welfare loss...**
 - arises from changes in consumption/production due to the introduction of a tax
 - depends on how much consumers and producers change their behavior
 - This is determined by the elasticities!
- **The greater the elasticity, the greater the welfare loss**
- **The greater the welfare loss, the more socially efficient (sales) purchases are not made**

Welfare Loss (Deadweight Loss - DWL)

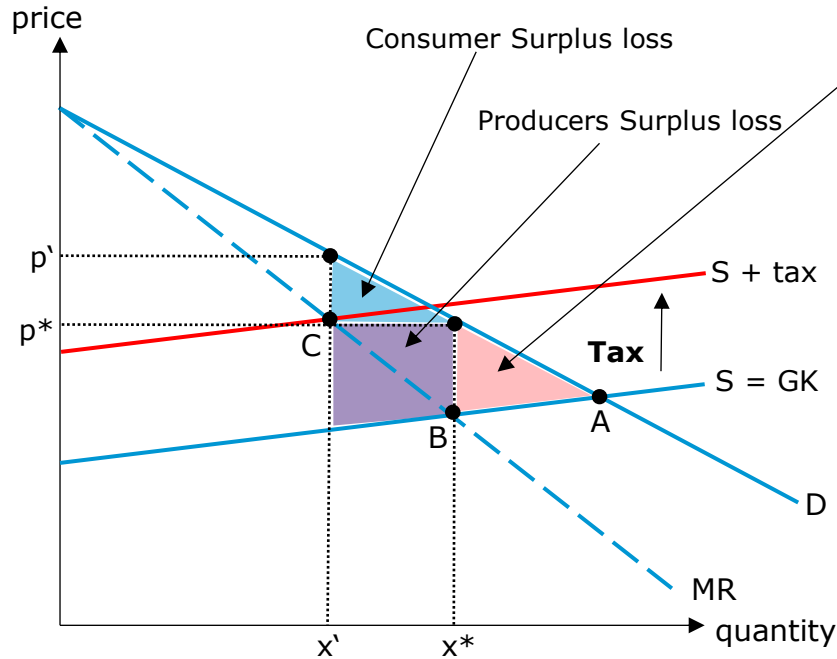
Marginal Welfare Loss



Quelle: Gruber, 5th edition, S. 627

- Marginal welfare loss increases with the tax rate
 - If the tax is increased by another \$0.10 from S_2 (equilibrium shifts from B to D), the welfare loss increases by DECB
- Taxes cause more welfare loss the further the equilibrium moves away from the tax-free equilibrium (cf. BCA with DECB)

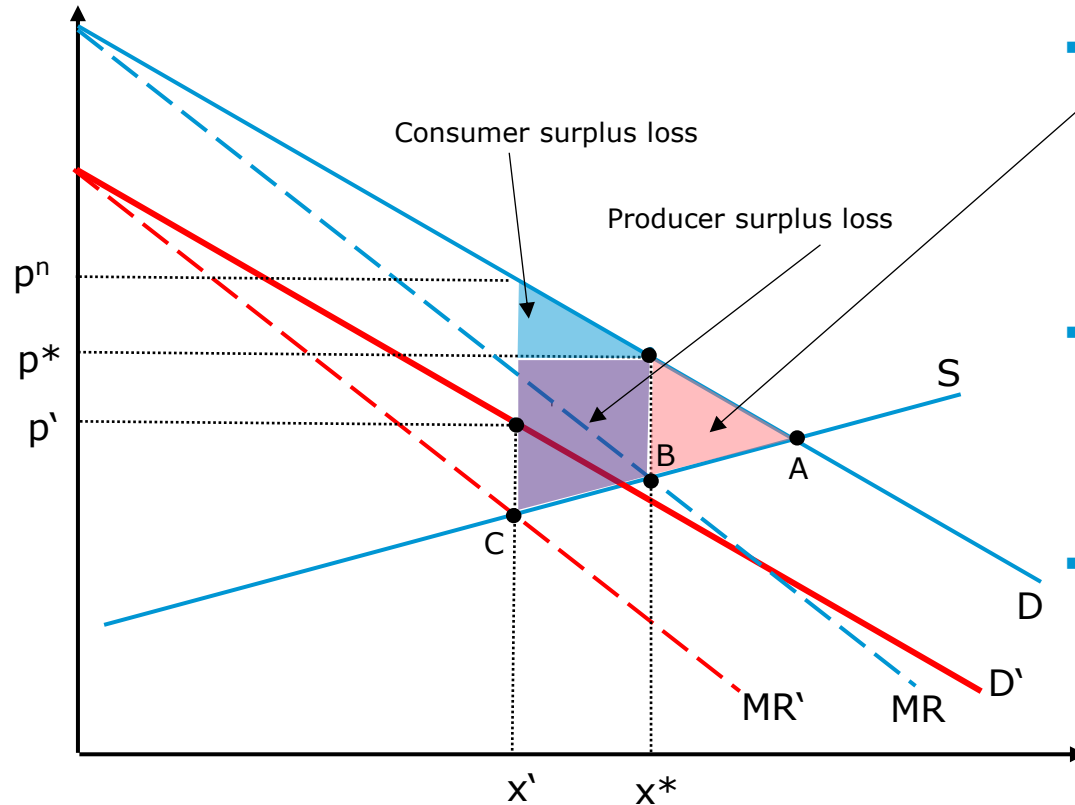
Welfare Loss (Deadweight Loss - DWL) Monopoly Loss



- **Monopoly Loss:** In a perfectly competitive market, equilibrium would be at point A → higher quantity (x), lower price (p). In an imperfect market, equilibrium is at point B with quantity x^* and price p^* .
- **Tax Introduction** (monopoly pays the tax, marginal cost curve shifts upward by the tax amount) → leads to a loss in economic rent. New equilibrium is at point C with quantity x' and price p' .
- **Marginal Welfare Loss**** is greater in imperfect markets because there is already a rent loss due to monopoly power (this is not counted as deadweight loss!).---Let me know if you'd like a diagram or a deeper explanation of these concepts!

Welfare Loss (Deadweight Loss - DWL)

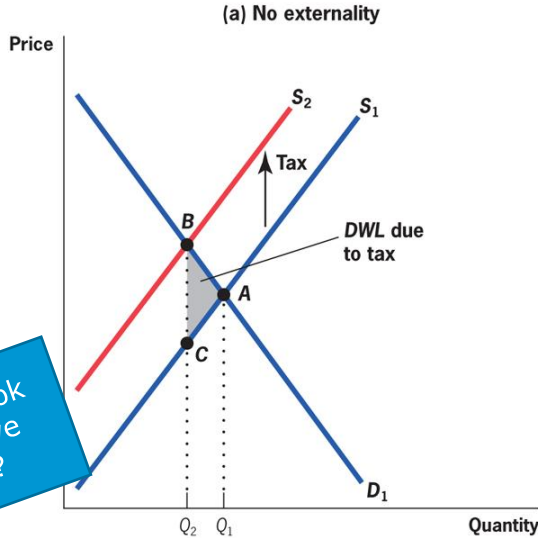
Monopoly Loss



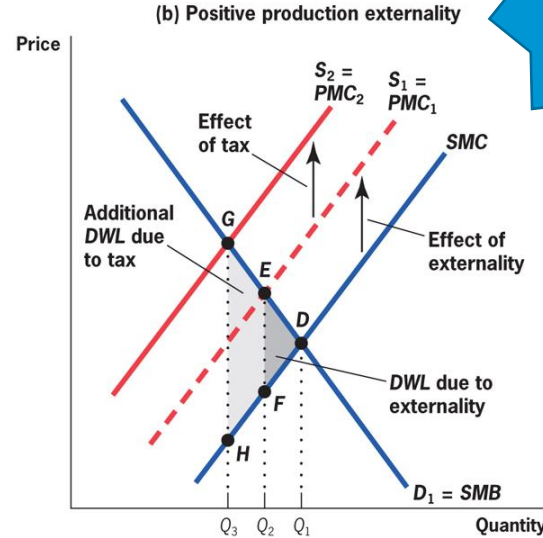
- **Monopoly loss:** in a perfectly competitive market, the equilibrium would be at A $\rightarrow$ more x , smaller p . In the imperfect market, we have the equilibrium at B with x^* , p^* .
- **Introduction of a tax** (monopoly pays tax, shift of GK up by the amount of the tax) $\rightarrow$ loss of surplus. New equilibrium at C with x' , p' .
- Marginal welfare loss is higher in imperfect markets because it includes the monopoly loss from the beginning (does not count towards DWL!)

Welfare Loss (Deadweight Loss - DWL)

Marginal welfare loss is higher in imperfect markets



How would it look with a negative externality?



Production uses individuals which are not paid for: $SMC < PMC$

- Due to a positive externality, even without a tax, less is produced than is socially optimal
- A tax reduces the traded quantity even further.
- The DWL of the same tax increase in a market with a positive externality (b) is much higher (GEFH) than in a market without one (a) (ABC).

- **Implications for the Efficiency of a Tax System**
 - Progressive tax rates can be less efficient → depends on redistribution
 - Welfare loss increases with the size of elasticities → tax goods that are as inelastic as possible
- **Marginal Welfare Loss**
 - Monopoly loss → preferably do not tax monopoly markets
 - It is better to tax many goods a little than a few goods a lot
 - Efficiency depends on existing distortions in the market (externalities, existing taxes)
- It is better to finance extraordinary expenses (e.g., wars, pandemics) with debt that is paid back over a long period of time through taxes

Deadweight loss- DWL

Example

HINT:
Sketch!!

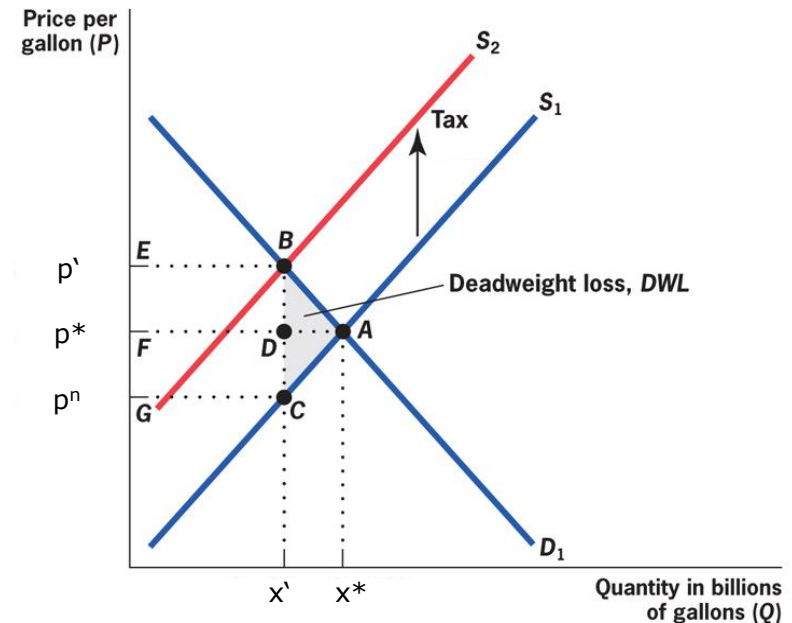
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ECONOMICS
AND BUSINESS

- The demand function in a market for a good is $p=1000-2x$ (x ... quantity and p ... price of the good). There is **perfect competition**. The marginal costs of production (= supply) are $MC=2x$. Fixed costs are zero.
- The state imposes a quantity tax with a tax amount of $t=30$ on the good, which must be paid by the producers.
- a. Calculate the price and quantity at equilibrium before and after the tax.
- b. Calculate the welfare loss (deadweight loss) that arises from the tax.
- c. Calculate what share of the welfare loss is borne by the producers and the consumers.

Deadweight loss- DWL

Example

- The demand function in a market for a good is $p=1000-2x$ (x ... quantity and p ... price of the good). There is **perfect competition**. The marginal costs of production (= supply) are $MC=2x$. Fixed costs are zero.
- The state imposes a quantity tax with a tax amount of $t=30$ on the good, which must be paid by the producers.
- a. Calculate the price and quantity at equilibrium before and after the tax.
- Before tax: Equilibrium A with x^* , p^*
- After tax: Equilibrium B with x' , p' and p^n



Deadweight loss- DWL

Example

Before Tax

- Equilibrium at A, $D=MC$

$$1000 - 2x = 2x$$

$$4x = 1000$$

$$\underline{x^* = 250}$$

$$p = 1000 - 2x \rightarrow \underline{p^* = 500}$$

After Tax

- Equilibrium at B, $D=MC+t$

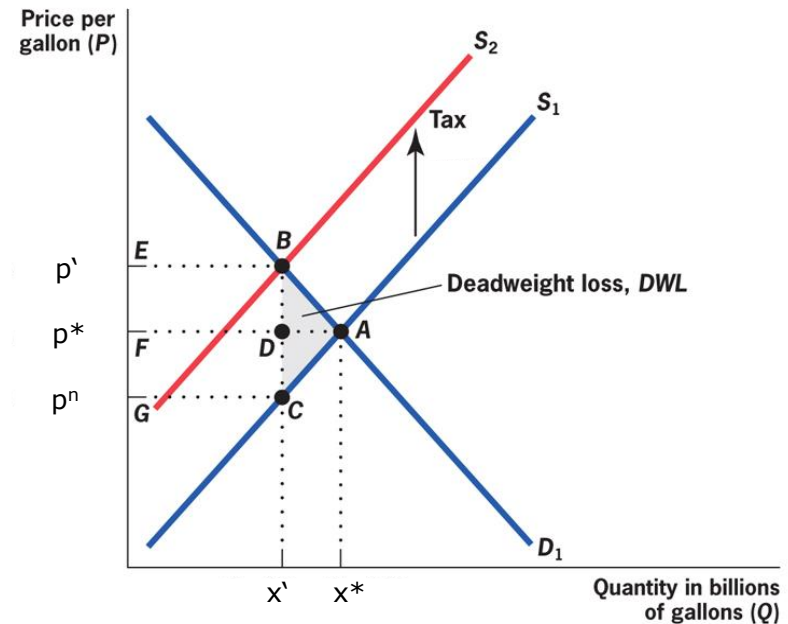
- $1000 - 2x = 2x + 30$

$$4x = 970$$

$$\underline{x' = 242,5}$$

$$\underline{p' = 515}$$

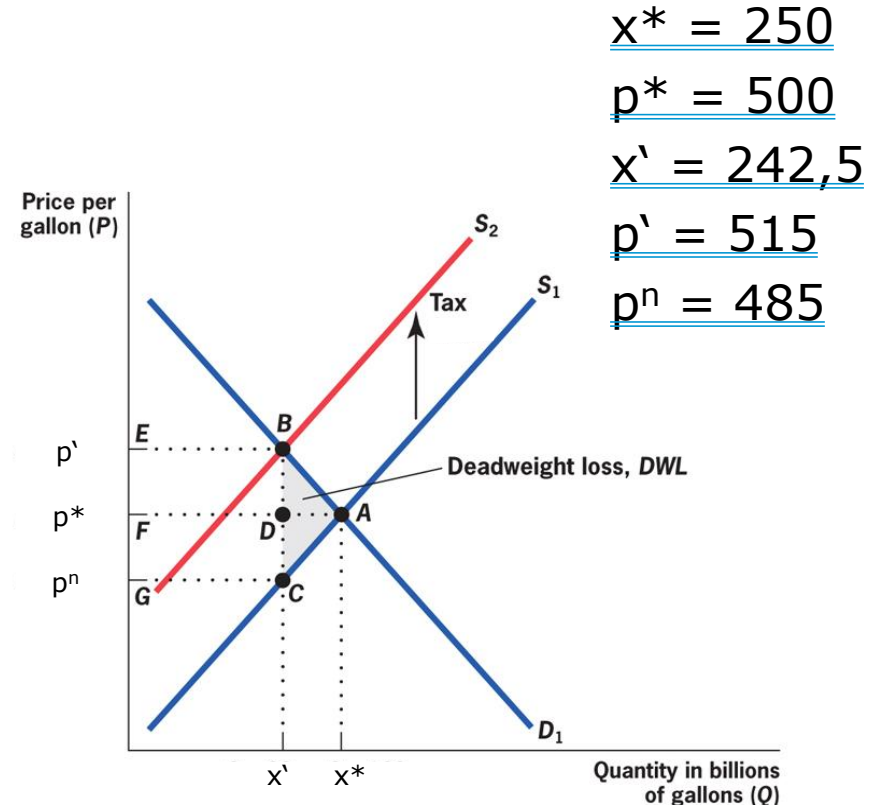
$$\underline{p^n = p' - t = 485}$$



Deadweight loss- DWL

Example

- The demand function in a market for a good is $p=1000-2x$ (x... quantity and p... price of the good). There is **perfect competition**. The marginal costs of production (= supply) are $MC=2x$. Fixed costs are zero.
- The state imposes a quantity tax with a tax amount of **t=30** on the good, which must be paid by the producers.
- a. Calculate the welfare loss (deadweight loss) that arises from the tax.
- Area of the triangle ABC
- b. Calculate what share of the welfare loss is borne by the producers and the consumers.
- K: ABD
- P: ACD



Deadweight loss- DWL

Example

a. Calculate welfare loss DWL

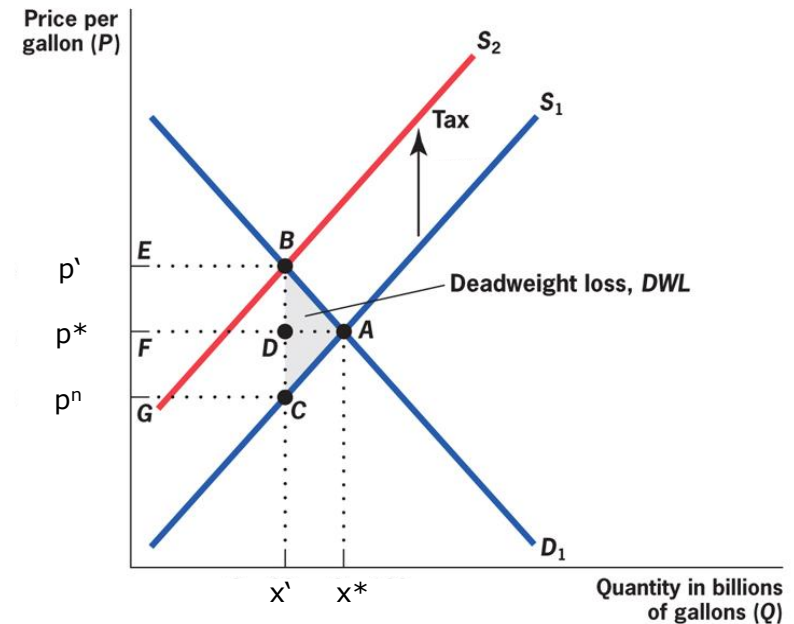
$$\begin{aligned} \text{(Area ABC): } & (x^* - x') \cdot t / 2 = (250 - 242,5) \cdot \\ & (515 - 485) / 2 = \underline{112,5} \end{aligned}$$

b. Distribution of the DWL:

$$\text{Konsument:innen (ABD): } (p' - p^*) \cdot (x^* - x') / 2 = \underline{56,25 \rightarrow 50\%}$$

$$\text{Produzent:innen (ACD): } (p^* - p^n) \cdot (x^* - x') / 2 = \underline{56,25 \rightarrow 50\%}$$

Note: not necessarily 50:50, should always be calculated separately!



Externalities



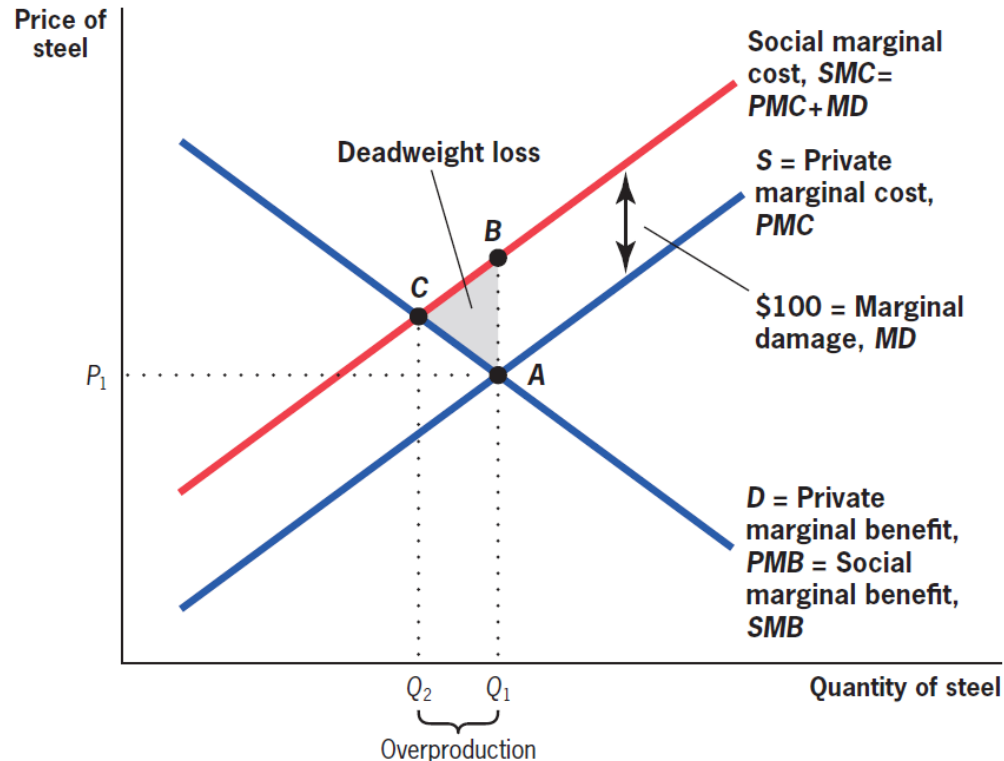
- **Perfectly Competitive Market:** The interests of firms and society are aligned (Invisible Hand)
- Marginal costs of production = social marginal costs
- Marginal benefits of production = social marginal benefits

- **Externalities** arise when the actions of individuals or firms have effects on other actors outside the market that are not "priced in".
- Can affect the supply or demand side.

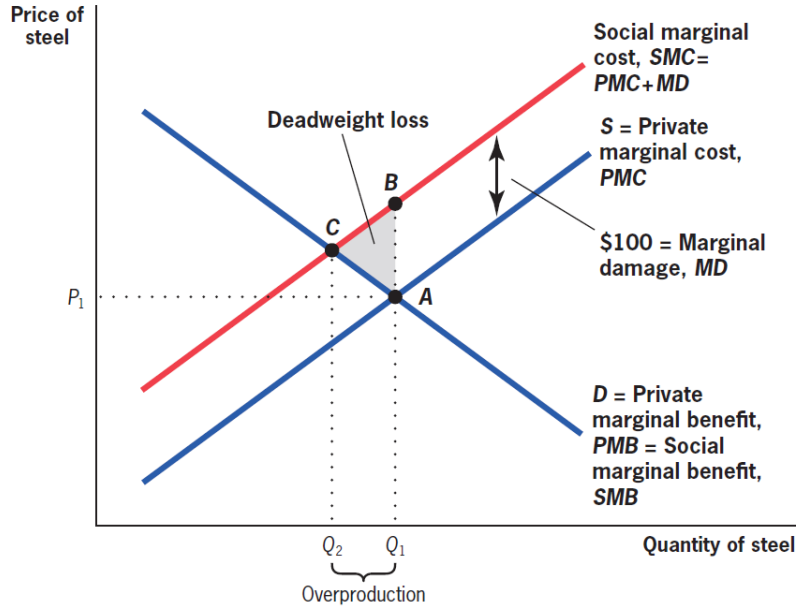
- **Negative Externalities:** others are negatively affected and are not compensated → Soc. Costs > Priv. Costs → too much Supply/Demand
- **Negative Production Externality:** When a company, through its production, harms others who are not compensated by the company.
 - A factory improperly disposes of toxic waste at a landfill, causing chemicals to seep into the ground.
- **Negative Consumption Externality:** When an individual's consumption reduces the well-being of others who are not compensated by the individual.
 - Someone smokes in a public space and thereby causes health damage to other people through passive smoking

Externalities

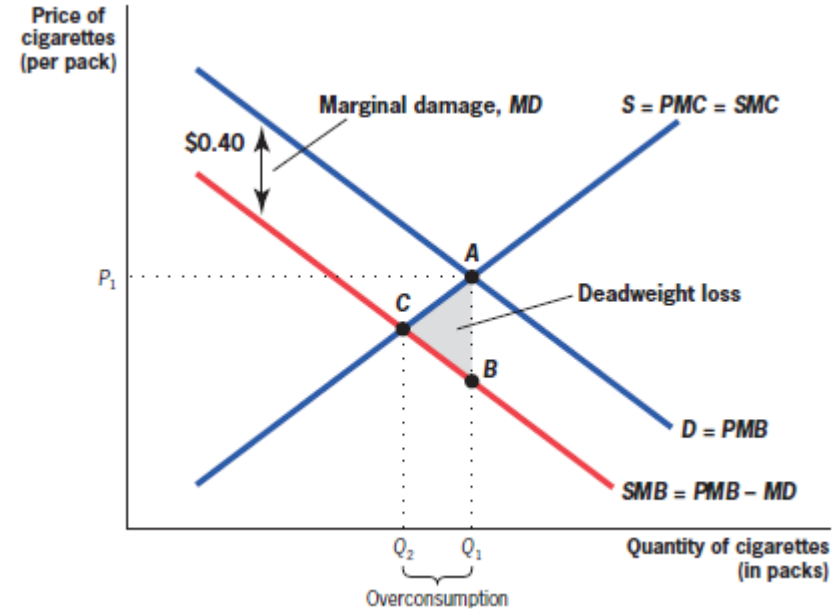
- Social Marginal Costs ($SMC = PMC + MD$):**
 The private marginal costs of producers plus damage **Private Marginal Costs (PMC):** The costs of an additional unit of production = supply curve **Marginal Damage (MD):** All additional costs associated with the production of the good but are not borne by the company
- Private Marginal Benefit (PMB):** The immediate benefit that results from the consumption of an additional unit of a good = demand curve **Social Marginal Benefit (SMB):** The private marginal benefit for the consumer minus the costs associated with the consumption of the good that are imposed on others



Externalities



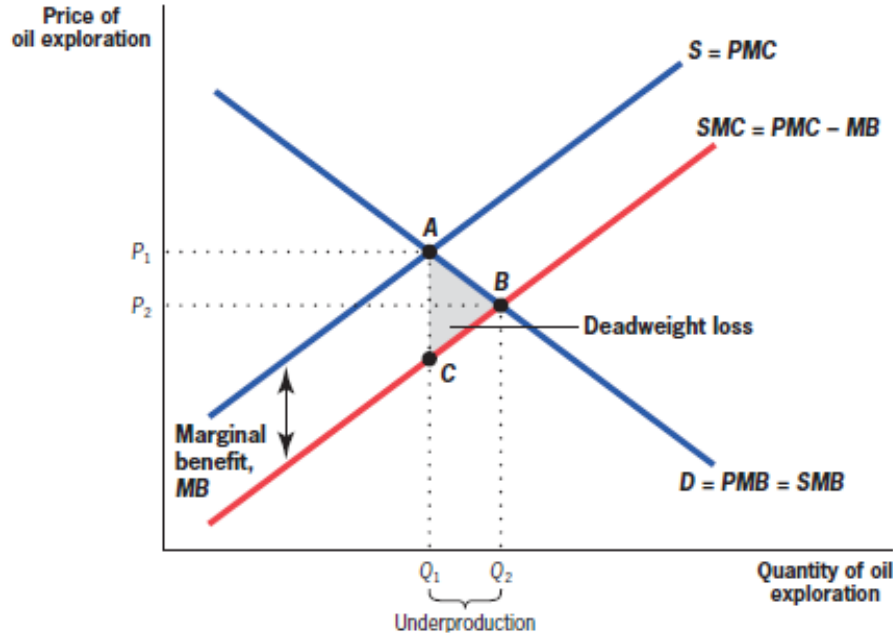
- **Negative Production Externality**
- Private marginal costs < social marginal costs
- Overproduction, Welfare loss



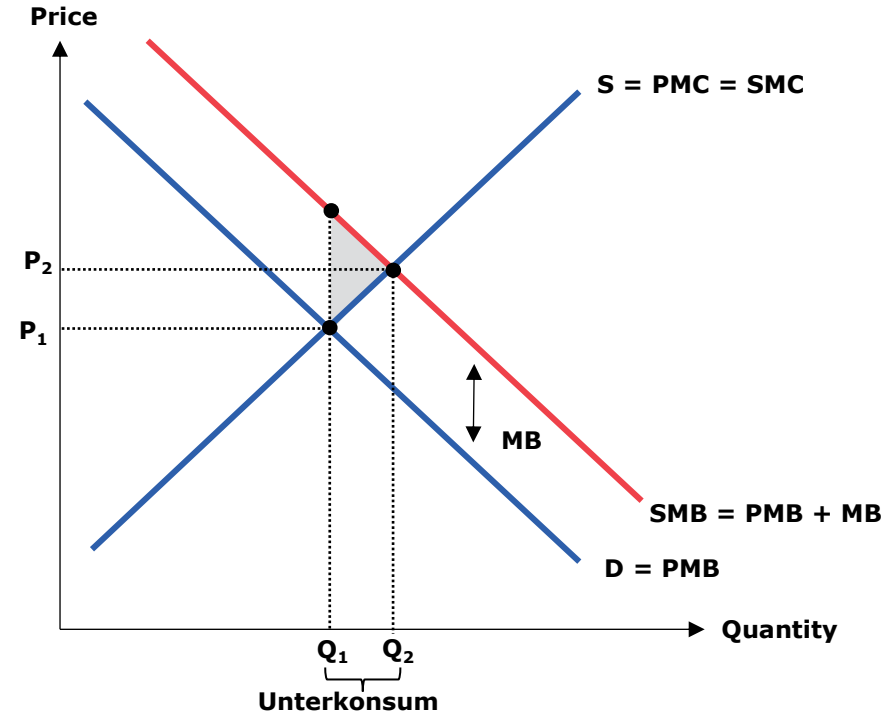
- **Negative Consumption Externality**
- Private marginal benefits > social marginal benefits
- Overconsumption, Welfare loss

- **Positive Externalities:** others benefit from actions and do not have to pay for them → Social Costs < Private Costs → too little Supply/Demand
- **Positive Production Externality:** When a company's production increases the well-being of others, but the company is not compensated by these others.
 - A company develops a new technology or medicine that can also be used by other companies or society without them having to bear the development costs.
- **Positive Consumption Externality:** When an individual's consumption increases the well-being of others, but the individual is not compensated by these others.
 - A person gets vaccinated and thereby protects not only themselves but also others from the spread of diseases (herd immunity).

Example



- **Positive Production Externality**
- Social marginal costs < private marginal costs
- Underproduction, Welfare loss



- **Positive Consumption Externality**
- Social marginal benefits > private marginal benefits
- Underconsumption, Welfare loss

- **Coase Theorem:** Market participants can internalize externalities themselves and achieve an efficient allocation of resources if:
- **Property rights** are clearly allocated (no matter to whom)
- **No transaction costs** exist
- **Complete information** is available
- But...
- **Imperfect information:** How high is the damage for whom? How high is the compensation?
- **Transaction costs:** "Bargaining is costly" (legal disputes) → inequalities
- **Property rights** = bargaining power

State-Based Solutions (Focus on Environment)

- **Market-based**
- **Fines and Taxes** to "internalize" external costs → "price" is set and quantity adjusts, e.g., CO₂ tax
 - Easier to implement than other measures
 - Generates tax revenue, questions about reinvestment!
 - How high? ONE price for everyone(s)?
 - Quantity reduction is not clear
 - Unexpected behavioral changes: it's socially acceptable to act in an environmentally harmful way if one "pays for it"?!
- **Subventions** to reduce external costs
 - Works only indirectly: does not internalize costs
 - Does not directly reduce harmful production, potentially provides incentives for redesign
 - Costs for taxpayers

State-Based Solutions (Focus on Environment)

- **Tradable (Emission) Certificates** → "quantity" is set and price adjusts
- Question of initial allocation ("giving away money")
- Lobbyism to increase certificates
- Must be adapted to the respective sector, high administrative effort
- Quantity is fixed, but no incentive to go below it (Ockenfels et al 2020)
- Price volatility → uncertainty

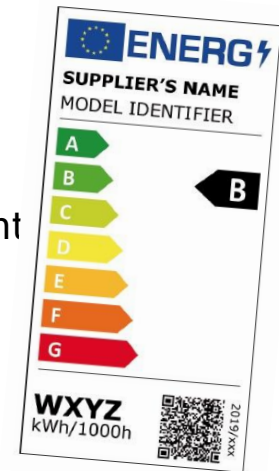
State-Based Solutions (Focus on Environment)

- **Regulations and Prohibitions**

- E.g., prohibitions, maximum limits for emissions
- High effort for control
- Potentially inefficient, because the same for everyone → less regressive
- Create "new environmentally friendly norms"

- **"Nudging"**

- Influencing behavior without prohibitions, commandments or economic incentive
- Subtle incentives to make certain decisions/products more attractive for individuals/consumers
- Only works for individual consumption, not e.g. for companies
- Criticism: "manipulation", who decides what is "right"?
- Examples: Default setting on the printer to "double-sided" printing, organic products at eye-level in the supermarket, "Veganuary" explanations, explanations on bins for waste separation, highlighting climate-friendly dishes on the menu

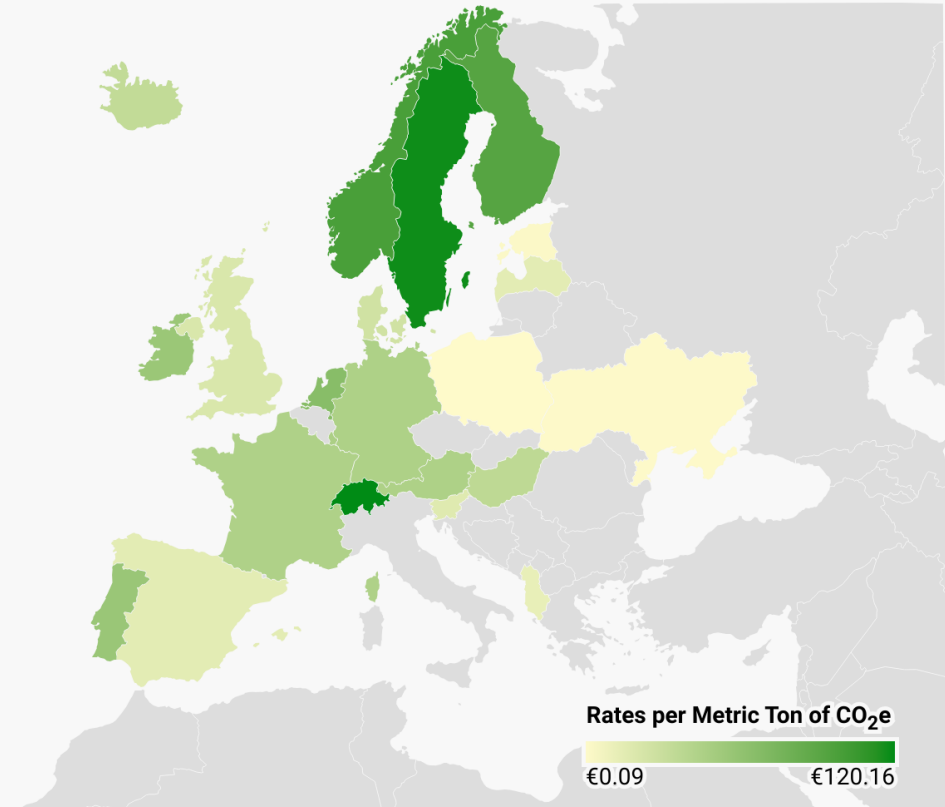


Carbon Pricing

Carbon Taxes in Europe

Carbon Tax Rates per Metric Ton of CO₂e, as of April 1, 2024

- The CO₂ price is mostly on mineral oils in the transport and building sectors: fuels, heating fuels like oil, gas
- How high is the price? Carbon Pricing Dashboard <https://carbonpricingdashboard.worldbank.org/compliance/instrument-detail>
- Ö: The price is being increased in stages, and is on average at the European level
- worldwide 2023 on average 23\$"
- To limit warming to 1.5°C, the global average price would have to rise to at least 170\$ by 2030" <https://systemschanlab.org/>



Note: The carbon tax rates were converted using the EUR-USD currency conversion rate as of April 1, 2024. Source: World Bank, "Carbon Pricing Dashboard", April 1, 2024; Ministry of Climate and the Environment, Poland, "Notice on Environmental Fee Rates for 2024", August 4, 2023; Ministry of Finance, Portugal, "Government Order No. 244-B/2023 (Article 2)", July 28, 2023.

- The EU's "Cap and Trade" system
- Introduced in 2005The world's first carbon market
- Heating fuels, electricity, air traffic, and since 2024 also maritime transport
- Covers a total of 45% of emissions
- 1t costs 68€ (as of 12/2024)
- Applies in all EU countries, Iceland, Liechtenstein, and Norway

EU Carbon Permits



source: tradingeconomics.com

Andersson (2019): "Carbon Taxes and CO₂ Emissions: Sweden as a Case Study" American Economic Journal: Economic Policy

Research Question: What was the effect of taxing fuel on emissions in Sweden?

- Introduction of a CO₂ tax and value-added tax on fuel
 - Price: 1991 US\$30 and rose to US\$132 per ton of CO₂ by 2018
 - Focus on the transport sector: ~40% of Sweden's CO₂ emissions
 - → Tax reform as a quasi-natural experiment

Synthetic control method

- World Bank annual panel data, 1960-2005
- "Synthetic Sweden" created from OECD countries: "How would emissions have developed in Sweden without the tax?"

Carbon Pricing

Empirical evidence

All Oecd states vs Sweden

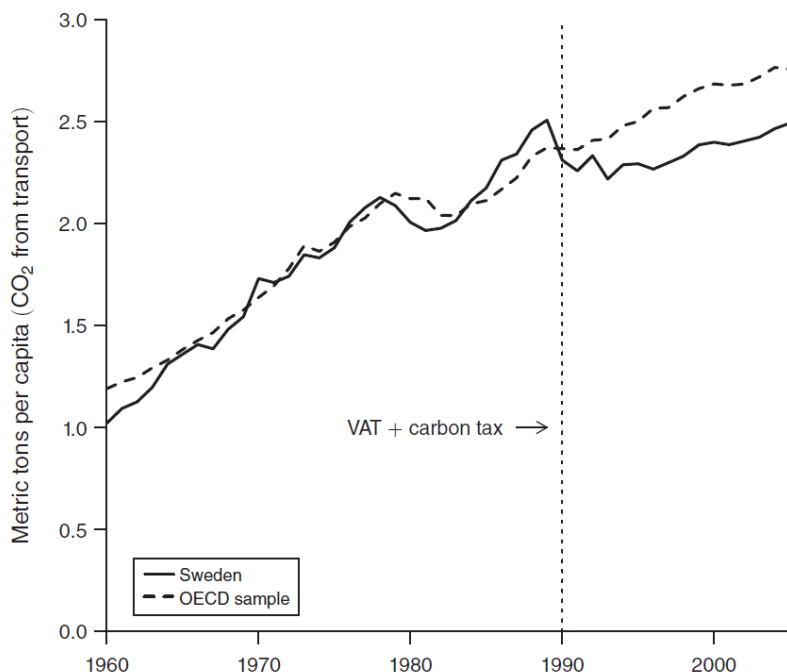


FIGURE 3. PATH PLOT OF PER CAPITA CO₂ EMISSIONS FROM TRANSPORT DURING 1960–2005: SWEDEN VERSUS THE OECD AVERAGE OF MY 14 DONOR COUNTRIES

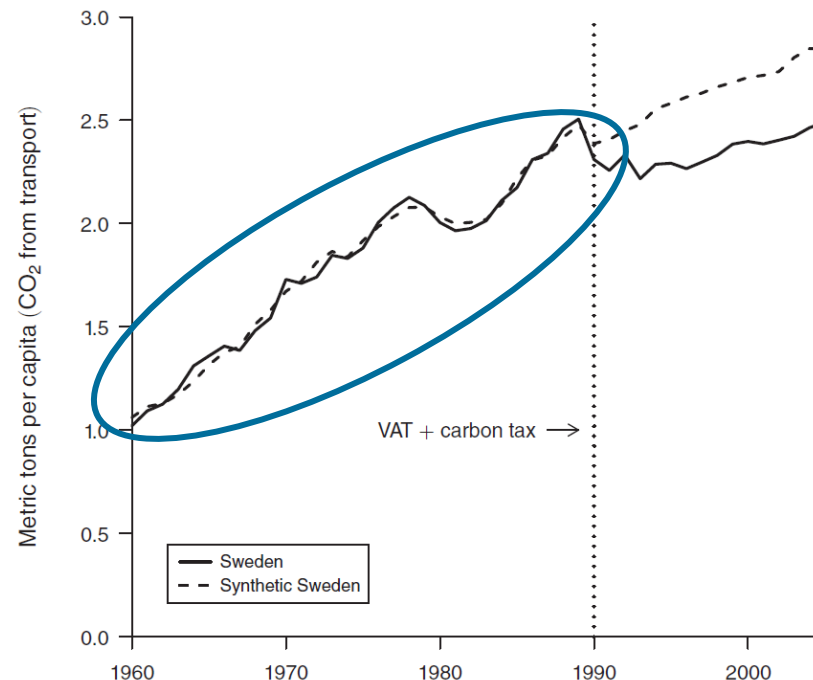
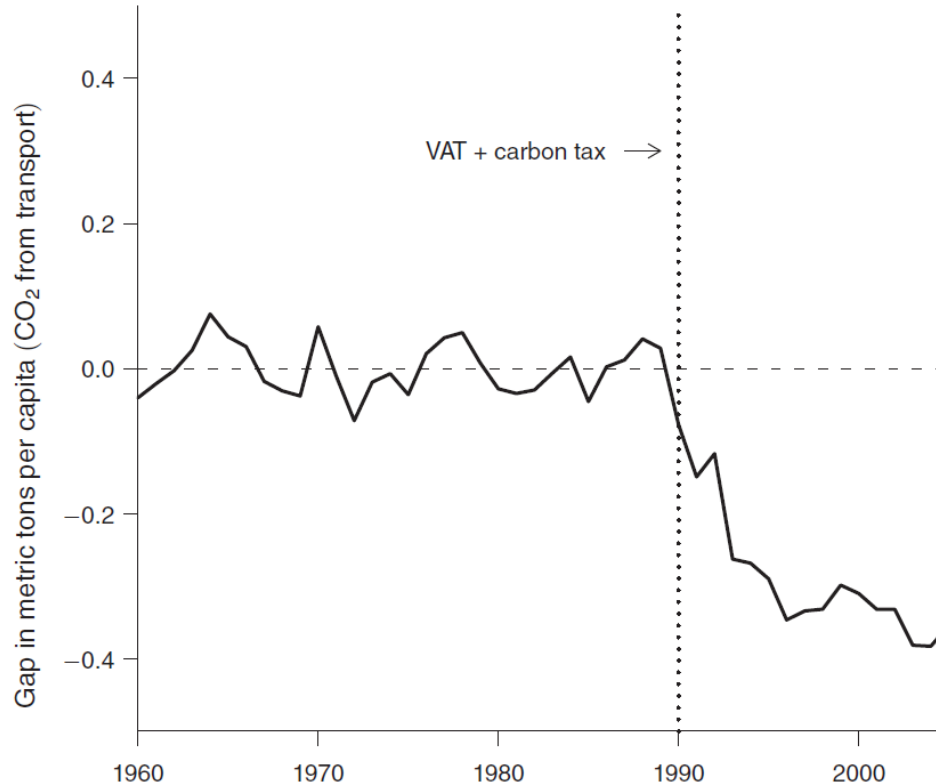


FIGURE 4. PATH PLOT OF PER CAPITA CO₂ EMISSIONS FROM TRANSPORT DURING 1960–2005: SWEDEN VERSUS SYNTHETIC SWEDEN

Carbon Pricing

Empirical evidence



- Reduction of emissions after tax introduction
- The effect gets bigger over time
- 2005: Sweden's emissions 12.5% below those of "synthetic Sweden"
- First time causal effects of a CO₂ price on emissions were shown

FIGURE 5. GAP IN PER CAPITA CO₂ EMISSIONS FROM TRANSPORT BETWEEN SWEDEN AND SYNTHETIC SWEDEN

Carbon Pricing

Empirical evidence

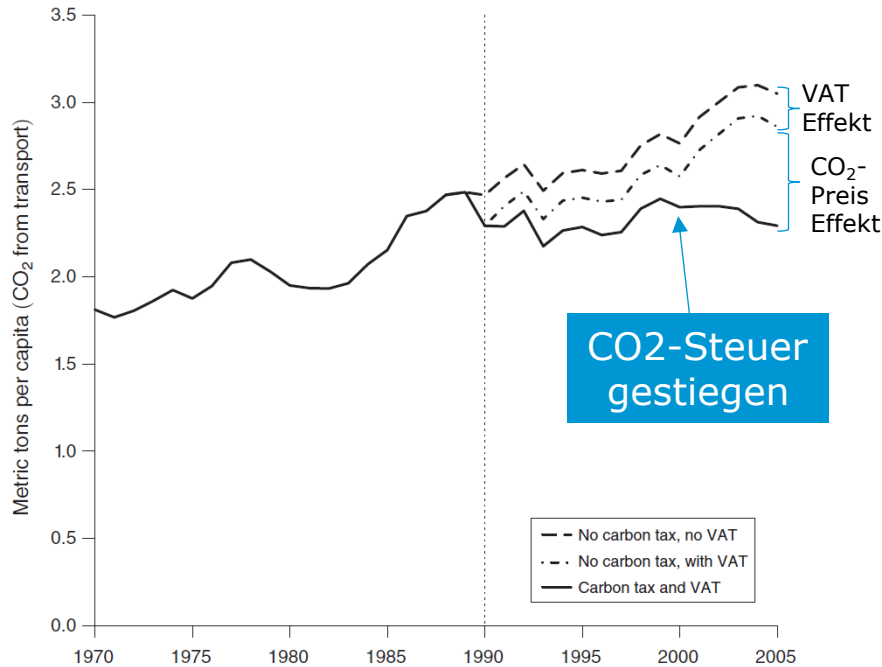


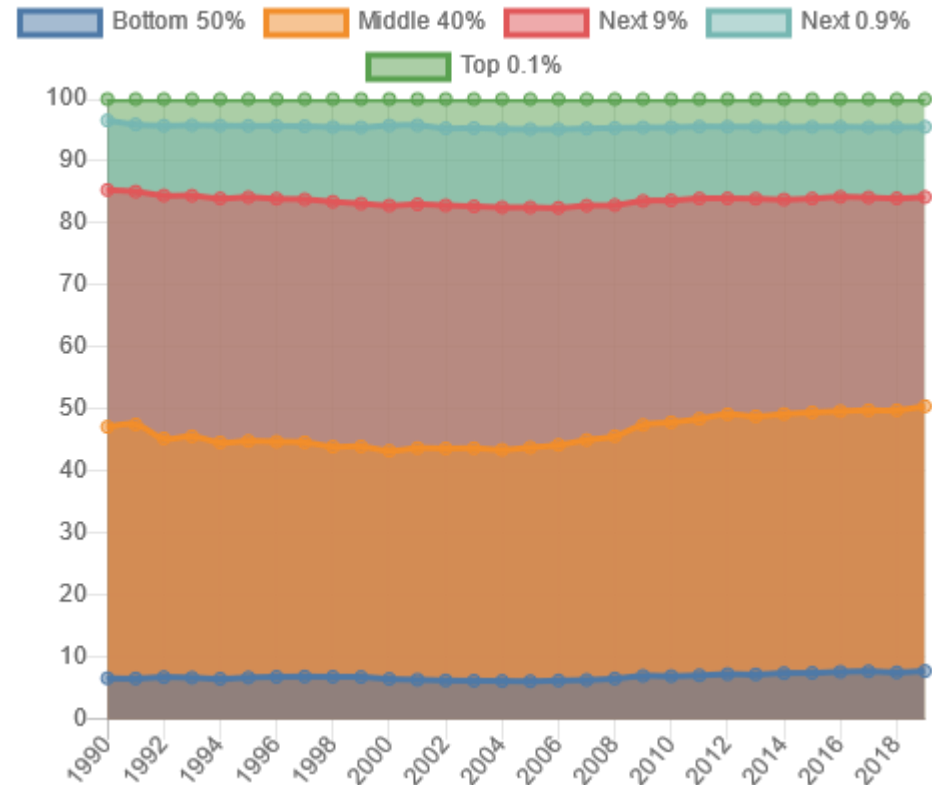
FIGURE 13. DISENTANGLING THE CARBON TAX AND VAT

- **Effects of each tax change:**
- Until 2020: VAT change and CO₂ tax contribute equally to emission reduction
- From 2020: Tax increase leads to further emission reduction
- Overall: Elasticity of the CO₂ tax > price elasticity
- Behavior changes more strongly due to the CO₂ price than due to other price increases

Environmental measures also have...

- ... Distributional effects!
- E.g., CO₂ taxes alone are regressive because they are consumption taxes: poorer households consume a larger portion of their income
- Carbon Inequality → Emissions increase with income!
 - Unequal distribution of CO₂ emissions between different groups: income, gender, ethnicity, caste, etc.

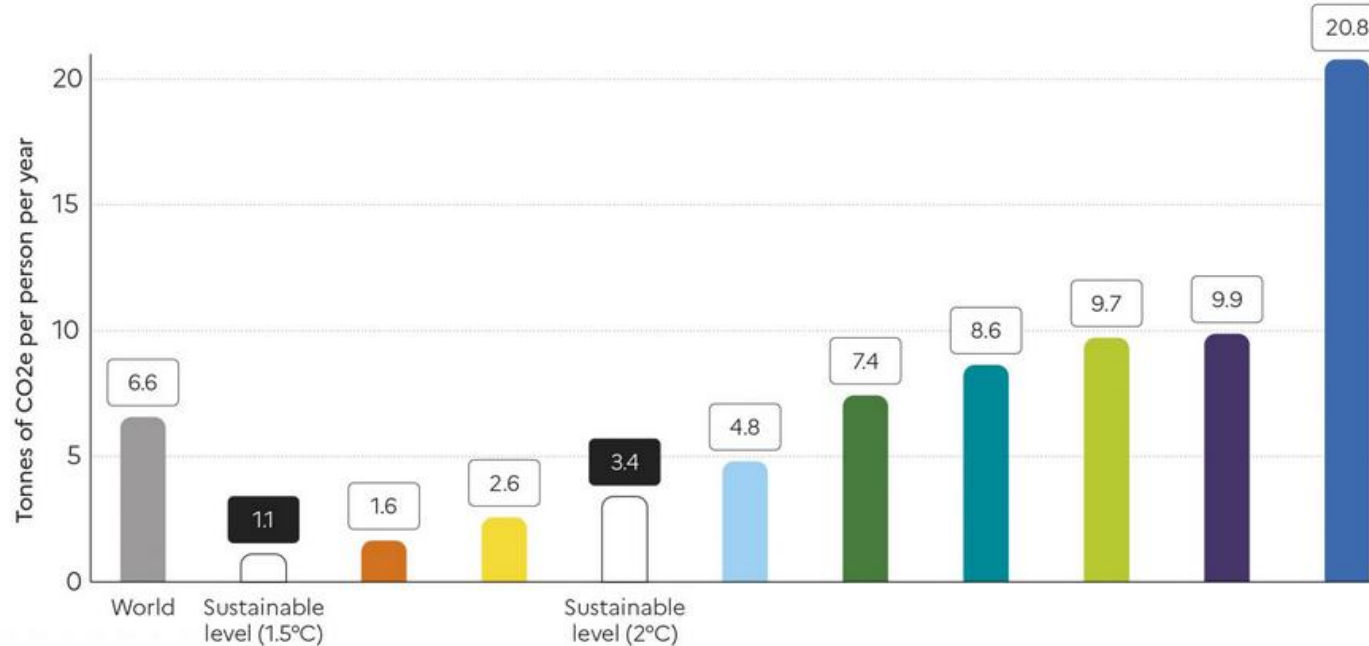
Emission Shares (%)



„Carbon Inequality“

Who has the most emissions?

Capita emissions by world region, 2019

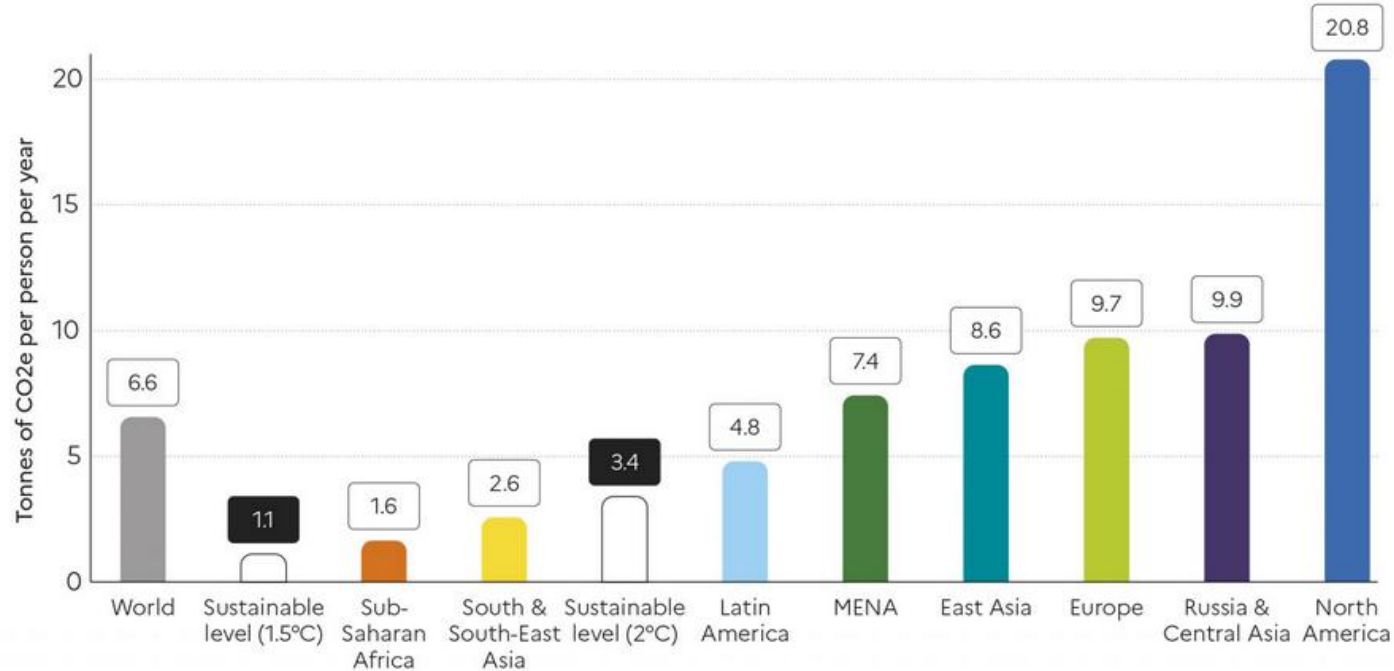


Interpretation: Values include emissions from domestic consumption, public and private investments as well as imports and exports of carbon embedded in goods and services traded with the rest of the world. Sustainable level correspond to an egalitarian distribution of the remaining carbon budget until 2050. **Sources and series:** [wir2022.wid.world/methodology](https://www.wir2022.wid.world/methodology) and Chancel (2021).

„Carbon Inequality“

Who has the most emissions?

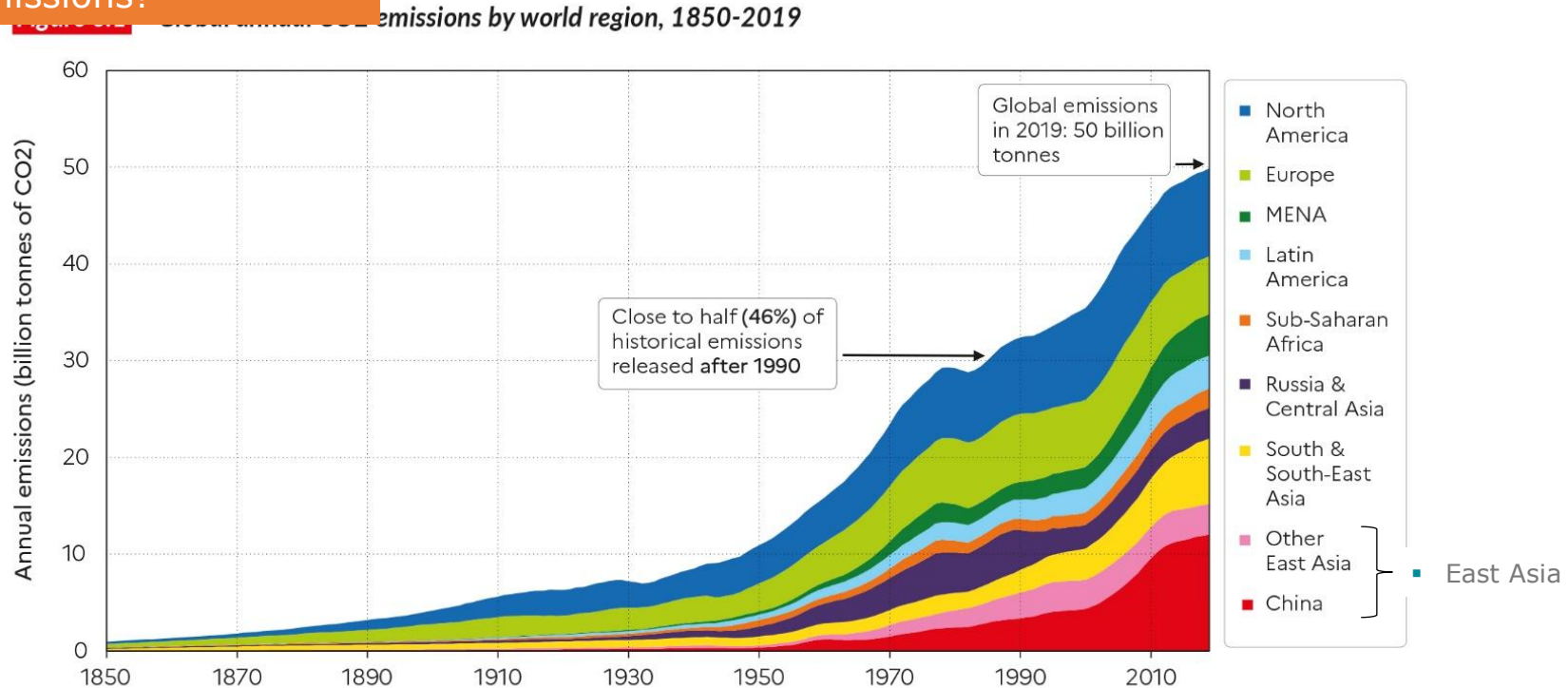
Capita emissions by world region, 2019



Interpretation: Values include emissions from domestic consumption, public and private investments as well as imports and exports of carbon embedded in goods and services traded with the rest of the world. Sustainable level correspond to an egalitarian distribution of the remaining carbon budget until 2050. **Sources and series:** wir2022.wid.world/methodology and Chancel (2021).

„Carbon Inequality“

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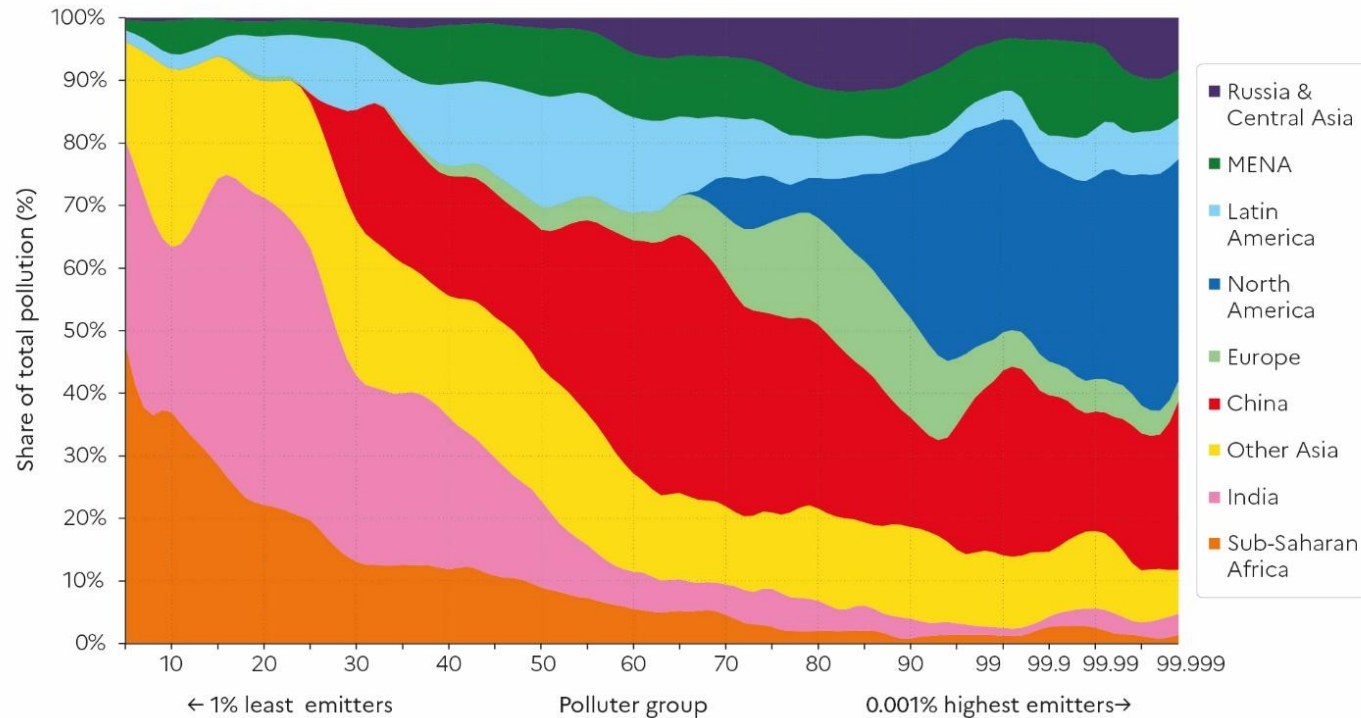


Interpretation: The graph shows annual global emissions by world regions. After 1990, emissions include carbon and other greenhouse gases embedded in imports/exports of goods and services from/to other regions. **Sources and series:** wir2022.wid.world/methodology and Chancel (2021). Historical data from the PRIMAP-hist dataset. Post-1990 data from Global Carbon Budget.

„Carbon Inequality“

Proportion of the population of each region along the "emissions distribution"

Figure 6.9a Geographical composition of global emitter groups, 2019



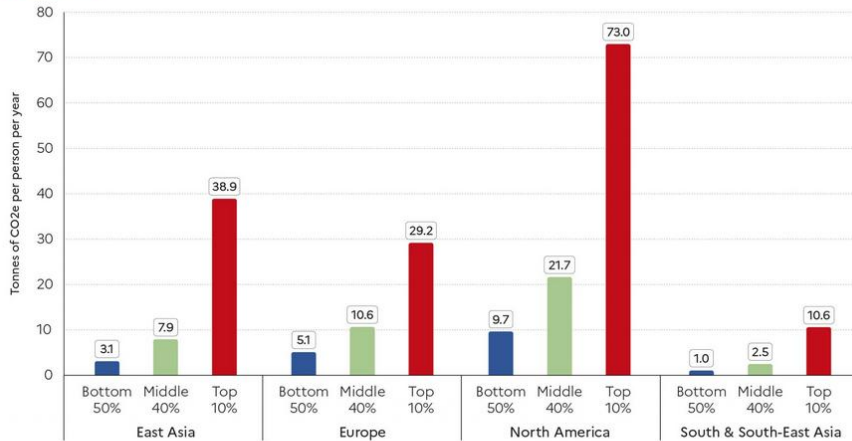
- China, Latin America, MENA represented along the "emissions distribution"
- Europe, North America represented in the upper half of the worldwide distribution → Discrepancy with high emissions
- High proportion of China in high emissions

Interpretation: The graph shows the share of world regions in each group of global emitters, from the bottom 1% to the top 0.001%.

Sources and series: [wir2022.wid.world/methodology](https://www.wir2022.wid.world/methodology) and Chancel (2021).

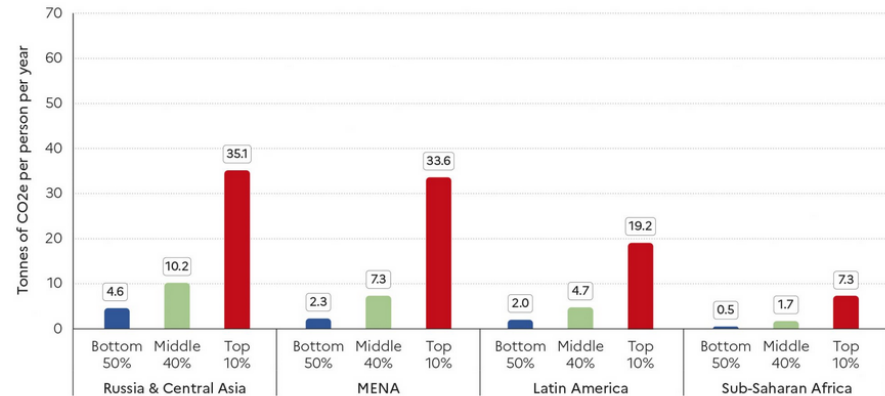
„Carbon Inequality“

Figure 6.4a Per capita emissions across the world, 2019



Interpretation: Personal carbon footprints include emissions from domestic consumption, public and private investments as well as imports and exports of carbon embedded in goods and services traded with the rest of the world. Modeled estimates based on the systematic combination of tax data, household surveys and input-output tables. Emissions split equally within households. **Sources and series:** wir2022.wid.world/methodology and Chancel (2021).

Figure 6.4b Per capita emissions across the world, 2019



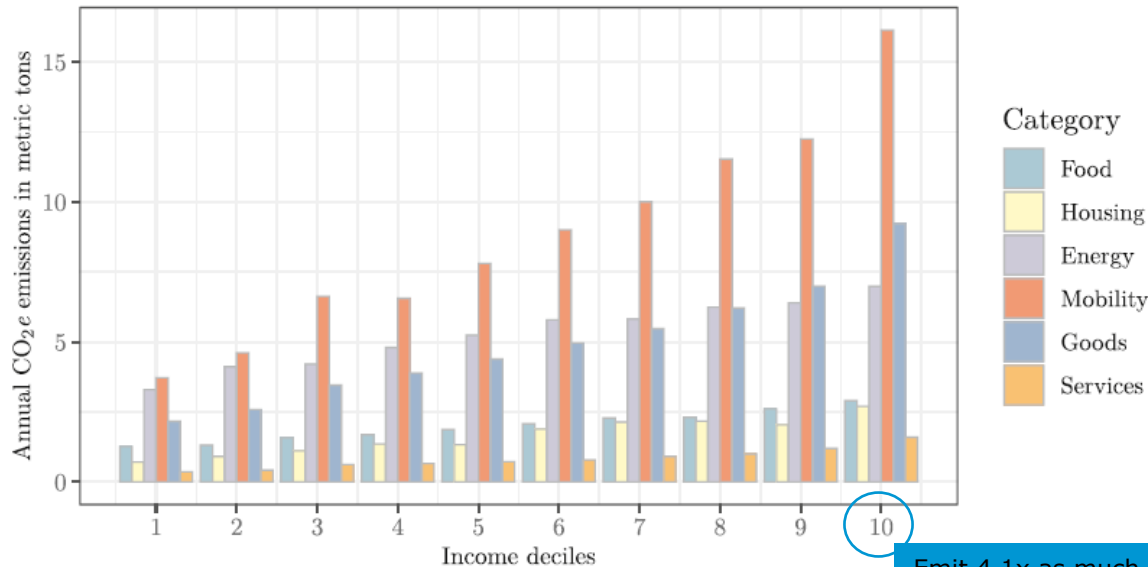
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World Inequality Report – Chapter 6 Carbon Inequality:
Learn@WU und [online](#)

„Carbon Inequality“

Empirical evidence

- Theine et al (2022): *"Emissions inequality: Disparities in income, expenditure, and the carbon footprint in Austria"* Ecological Economics
- They analyze the distribution of emissions along the income distribution
→ clear positive increase

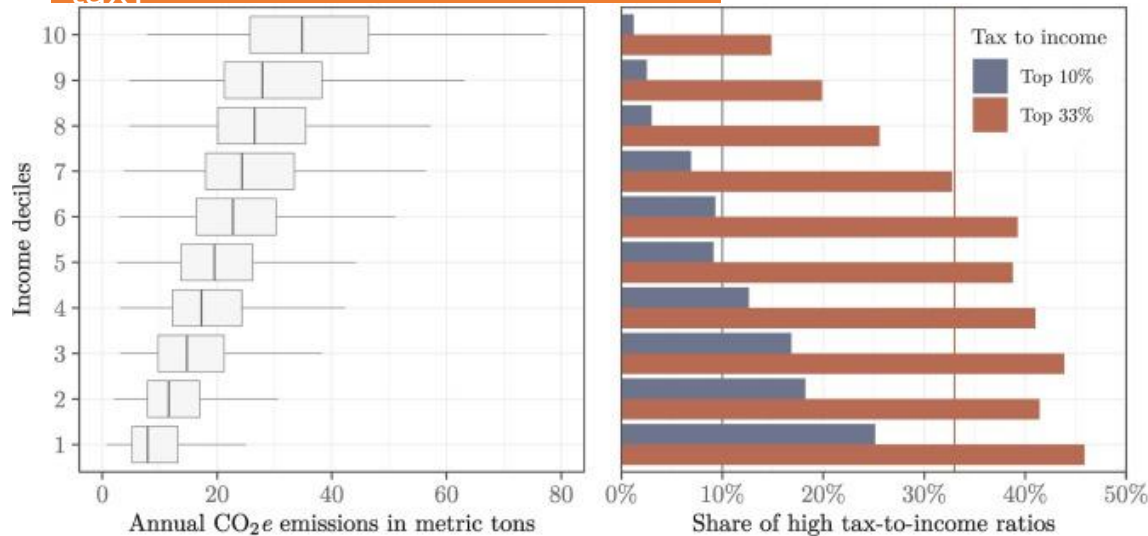


Emit 4.1x as much as
the lowest decile

„Carbon Inequality“

Empirical evidence

And who is affected by the CO₂ tax?



- While the 1st decile emits the least (left), a large part of these people are in the top 10% and 33% of those who pay the most CO₂ tax relative to their income (right)
- In general: Lower income deciles pay more CO₂ tax relative to their income

„Carbon Inequality“



- We have seen:
- There are strong differences between regions in who emits how much
- Within regions/countries there are strong differences in who emits how much
- A tax hits certain income groups harder - those who emit less
- **How can we counteract these distributional effects?**

- **What does eco-social mean?**
- **Eco:** Aims at improving the ecological situation
- **Social:** Redistribution from the upper to the lower income groups

- **Why eco-social measures?**
- Interaction between climate and social policy
- Cost-intensive climate policy can use up funds for social policy
- Regressive effects, "eco-poverty"? → Social inequalities can be widened or exacerbated
- SOURCE: GUGUSHVILI AND OTTO (2021)

- **Which income group should be addressed by the measures?**
- **Lower End of the Income Distribution**
 - Well-being of the poor
- **Middle Class**
 - Well-being of the middle class, who had access to good jobs but are now possibly confronted with a shortage of income/assets and greater uncertainty?
- **Top Incomes/Wealth**
 - Do we question the concentration of economic power and the associated political influence of wealthy individuals and large corporations?
- Our political priorities influence whether we target the lower, middle, or upper end of the income distribution.

- Which income group can be addressed by which measures?
- Think of a few examples!



Einkommens-/Vermögensverteilung

- **Lower 50%**
- **Increasing the Supply of Green Energy**
- Industrial policy: Public investment in renewable energy
- Social protection: Increased transfers for workers in industries affected by restructuring
- **Increasing Access to Green Energy**
- Public investment in access to green energy
- Building new CO₂-neutral social housing
- **Changing Energy Use (Buildings, Transport, Industry)**
- Expansion of public transport
- Energy-efficient renovation of social housing
- Transfer payments to offset price increases from the CO₂ price

- **Eco-Social Measures**
- **Middle 40%**
- **Increasing the Supply of Green Energy**
- Measures for the lower 50%
- Financial incentives to encourage middle-class investment in green energy
- Prohibitions on new investments in fossil fuels
- **Increasing Access to Green Energy**
- Subsidies for the construction of energy-efficient residential buildings
- Building codes
- Fines and prohibitions on the sale of inefficient residential buildings
- **Changing Energy Use (Buildings, Transport, Industry)**
- Measures for the lower 50%
- Stricter regulations and taxes on environmentally harmful products (e.g., SUVs, flight tickets)
- Subsidies for green alternatives (e.g., e-cars)

- **Top 10% and 1%**
- **Increasing the Supply of Green Energy**
- Wealth and corporate taxes with environmental surcharges to finance environmental measures
- Acceleration of the phase-out of fossil fuels
- Prohibitions on new investments in fossil fuels
- **Increasing Access to Green Energy**
- Wealth or corporate taxes with environmental surcharges
- Abolition of subsidies for fossil fuels
- **Changing Energy Use (Buildings, Transport, Industry)**
- Strict regulations for environmentally harmful consumption (e.g., SUVs, flight tickets)
- Wealth and corporate taxes with environmental surcharges
- Introduction of CO₂ accounts for monitoring and limiting high personal CO₂ footprints

		What kind of climate policy?		
		Increase green energy supply	Increase green energy access	Switch in energy end-uses (building, transport, industry)
Which social group is targetted?	Bottom 50%	Industrial policy: public investments in renewables (off or on-grid); Social protection: increase transfers to workers in industries affected by the transition	Public investments in green energy access (e.g. clean cookstoves; construction of new zero carbon social housing)	Develop public transport systems: low-carbon bus, rail, car-sharing strategies; energy retrofitting in social housing; cash-transfers to compensate increase in fossil energy prices
	Middle 40%	Same as above + Financial incentives to encourage middle-class investments in green energy. Bans on new fossil investments	Subsidies for green housing construction; Buildings regulations; penalty and bans on sales of inefficient housing	Same as above; Stricter regulations & taxes on polluting purchases (SUVs, air tickets); Subsidies on green alternatives (elec. vehicles)
	Top 10 % & Top 1%	Wealth or corporate taxes with pollution top-up to finance the above & accelerate divestment from fossils; Bans on new fossil investments	Wealth or corporate taxes with pollution top-up (see left); Fossil fuel subsidy removal*	Strict regulations on polluting purchases (SUVs, air tickets); Wealth or corporate taxes with pollution top-up (see left); Carbon cards to track high personal carbon footprints & cap them

World Inequality Report – Chapter 6 Carbon Inequality: [online](#)

References



References and further links

- [EUROSTAT: EU-SILC](#)
- [help.gv.at -> Steuerstufen](#)
- [Statistik Austria](#)
- [Das Steuerbuch 2018](#)
- Falter Interview: Privat oder Staat (auf Canvas)
- Adda, J., & Cornaglia, F. (2006). Taxes, cigarette consumption, and smoking intensity. *American Economic Review*, 96(4), 1013-1028.
- [Benzarti \(2024\): "Tax Incidence Anomalies"](#). Prepared for the Annual Review of Economics. NBER Working Paper No. 32819.
- Chetty, R., Looney, A., & Kroft, K. (2009). Salience and taxation: Theory and evidence. *American economic review*, 99(4), 1145-77.
- Douenne, T., & Fabre, A. (2022). Yellow vests, pessimistic beliefs, and carbon tax aversion. *American Economic Journal: Economic Policy*, 14(1), 81-110.
- Oates, W. E. & Schwab R. M. (2015). The Window Tax: A Case Study in Excess Burden. *Journal of Economic Perspectives*, 29(1) S. 163 – 180 ([Link](#))
- Ockenfels et al (2020) Pricing Externalities and Moral Behaviour. In: *Nature Sustainability* 3, 872–877
- Silvia Rocha-Akis & Jürgen Bierbaumer-Polly & Julia Bock-Schappelwein & Martina Einsiedl & Michael Klien & Thomas Leoni & Simon Loretz & Hedwig Lutz & Christine Mayrhuber, 2019. "Umverteilung durch den Staat in Österreich 2015," WIFO Studies, WIFO, number 61782, August.
- Theine, H., Humer, S., Moser, M., & Schnetzer, M. (2022). Emissions inequality: Disparities in income, expenditure, and the carbon footprint in Austria. *Ecological Economics*, 197, 107435.
- World Inequality Report – Chapter 6 Carbon Inequality
- <https://taxfoundation.org/carbon-taxes-in-europe-2021/>
- https://carbonpricingdashboard.worldbank.org/map_data